CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 13: Section and Mid-Point Formula

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the point that divides the segment joining $(1, 3)$ and $(6, 8)$ in the ratio $2 : 3$.
2. Find the mid-point of the segment joining $(-5, 4)$ and $(3, -6)$.
3. In what ratio does the point $(1, 1)$ divide the segment joining $(-2, -5)$ and $(3, 5)$?
4. In what ratio does the x-axis divide the segment joining $(3, -4)$ and $(8, 6)$? Find the point.
5. Find the points of trisection of the segment joining $(0, 3)$ and $(9, 9)$.
6. The mid-point of AB is $(2, -1)$ and $A = (6, 3)$. Find B.
7. $A(1, 1)$, $B(5, 2)$ and $C(7, 6)$ are three vertices of a parallelogram $ABCD$. Find D.
8. Find the centroid of the triangle with vertices $(3, 5)$, $(-1, 2)$ and $(4, -1)$.
9. The centroid of a triangle is the origin and two of its vertices are $(4, -3)$ and $(-1, 5)$. Find the third vertex.
10. $A(2, 6)$ and $B(8, 2)$ are the ends of a diameter of a circle. Find its centre and its radius.

MathSetu

Answer Key

1. $(3, 5)$
2. $(-1, -1)$
3. $3 : 2$
4. $2 : 3$; $(5, 0)$
5. $(3, 5)$ and $(6, 7)$
6. $(-2, -5)$
7. $(3, 5)$
8. $(2, 2)$
9. $(-3, -2)$
10. Centre $(5, 4)$; radius $\sqrt{9 + 4} = \sqrt{13}$ units

MathSetu